4 The sum $\frac{1}{2!}+\frac{2}{3!}+\frac{3}{4!}+\cdots+\frac{2021}{2022!}$ can be expressed as $a-\frac{1}{b!}$, where a and b are positive integers. What is $a+b$? (2014 AMC 10A Problems, Question #9)

A. 2020 B. 2021 C. 2022 D. 2023 E. 2024

6 Calculate: $\frac{4}{3\times 7}+\frac{4}{7\times 11}+\frac{1}{11\times 12}+\frac{3}{12\times 15}+\frac{2}{15\times 17}+\frac{12}{17\times 29}+\frac{1}{29}=$ ______ .

7 Calculate: $\frac{1}{3}+\frac{1}{15}+\frac{1}{35}+\frac{1}{63}+\frac{1}{99}+\frac{1}{143}=$ ______ .

5 If a four-digit number $\overline{2X9Y} = 2^X \cdot 9^Y$, then $X =$ ______ and $Y =$ ______ .

7 Among the natural numbers less than 5000, how many numbers are divisible by 11 and have a digit sum of 13?

A. 15 B. 18 C. 21 D. 24

8 A four-digit number with each digit different can be divisible by $1, 2, 3, 4, 5, 6, 7$. What is the greatest such number? ______

Lesson 3

10. The least common multiple of x and y is 60. The least common multiple of y and z is 70. The least common multiple of x and z is 84. Then $x =$ ______ .

A. 6 B. 10 C. 12 D. 15 E. 20

Extention

2. The sum of the greatest common divisor and the least common multiple of 12, 18, and 90 is ______ .

9. Let $a@b = [a, b] + (a, b)$, where $[a, b]$ represents the least common multiple of a and b, and (a, b) represents the greatest common divisor of a and b. Given $12@x = 52$, then $x =$ ______ .

If x and y satisfy the system of linear equations $\begin{cases} 323x + 457y = 1103 \\ 177x + 543y = 897 \end{cases}$, find the value of $x^4 + 4x^2y^2 + 5y^4$.

5 Solve the following system of linear equations:

$$\begin{cases} x_1 + x_2 + x_3 = 1 \\ x_2 + x_3 + x_4 = 2 \\ x_3 + x_4 + x_5 = 3 \\ x_4 + x_5 + x_1 = 4 \\ x_5 + x_1 + x_2 = 5 \end{cases}$$

❸ The equation $a(3x-2)+b(2x-3)=8x-7$ (with respect to x) has infinitely many solutions, then $a =$ ______ .

❺ Solve the equation with respect to x: $2ax=(a+1)x+b$.

❹ Solve the following equations with respect to x:

(2) $m(x + n + 3) = m(n + 3) + n + 9$

❸ Solve the following equations with respect to x:

(1) $(2a - 1)x = a + 1$

(2) $3ax - 2 = a - 6x$

❾ Solve the following system of linear equations:

$$\begin{cases} |x| + y = 12 \\ x + |y| = 6 \end{cases}$$

Lesson 5

1 Solve $|x-2| \leqslant 2x-10$.

2 Solve $|2x-4|+|3x-3| \leqslant 5$.

Lesson 5 HW

5 Solve the following system of linear inequalities $\begin{cases} \dfrac{x-3}{2}+3>x+1 \\ 1-3(x-1) \leqslant 8-x \end{cases}$, and graph the solution.

8 Solve the absolute value inequality $|x - 2| > 2x + 1$, obtaining $x <$ ______ .

Lesson 5 Ext

10 Solve the following absolute-value inequalities

(1) $|3x + 1| > 2x$ (

5 Solve the following inequalities:

(1) $|2x - 3| \geqslant x + 1$

⑩ Solve the following absolute-value inequalities:

(1) $|2x + 1| - |5 - x| > 2$

(2) $|x - 1| + |x + 1| < 9$

L6

4 There is an equation $y = (m-1)x^{|m|} + n - 4$.

(1) If this is a linear equation, what is the value of m and n?

L7

8. If the graph of the line $y = 2x - 1$ is rotated 90° counterclockwise about the origin O, the equation of the line after the rotation is ______ .

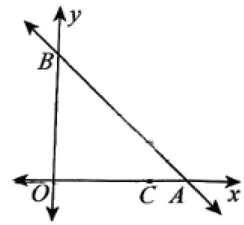
L7 Extention

5 The line that is symmetric to line $y = mx - 3$ about the x-axis passes through $(-1, 2)$, then the value of m is _____ .

A. 1 B. -1 C. 5 D. -5

8 The equation of the line obtained by rotating line $y = 3x$ clockwise around the origin $90°$ is _______ ; The equation of the line obtained by rotating line $y = 3x$ counterclockwise around the origin $90°$ is _____

10 The graph of the linear equation $y = -x + 5$ intersects with x-axis and y-axis at A and B respectively .



(3) Points M and N are on the line AB and y-axis respectively, so that the perimeter of $\triangle CMN$ is at its minimum. What are the coordinates of point N?

(2) $a^2 + b^2 + c^2 + ab + bc - ca =$ ______ .

(3) $a^2 + b^2 + c^2 - ab + bc - ca =$ ______ .

3 Calculate the following:

(1) $(2 + 1)(2^2 + 1)(2^4 + 1) \cdots (2^{2^n} + 1) =$ ______ .

(3) $(2a+b)^2\left[4a^2-(2a-b)b\right]^2 =$ ______

L8 Extention

6 **Calculate:**

(1) $(2y-x-3z)(-x-2y-3z) =$ ______ .

Lesson 9

2 Factorize: $x^2(a+b)^2 - 2xy\left(a^2 - b^2\right) + y^2(a-b)^2$

L9 HW

7 Factorize the following expressions:

(4) $-20x^2 + 9x + 20 =$ ______ .

L9 Extention

3 Factorize: $-3abx^4 + acx^3 - ax =$ ______ .

5 Factorize: $x^2 + 144y^2 - 25xy =$ ______ .

4 Factorize: $(2x - 3y)(3x - 2y) + (2y - 3x)(2x + 3y) - (6x - 4y)(6x - 9y) =$ ______ .

8 Factorize: $x^2 - (a + \frac{1}{a})x + 1 =$ ______ .

Lesson 10

3. As shown in the picture, there are three cards, each with a number on it. If we pick one or more cards from them (we may not rotate the cards) to form some numbers, we can get _____ different prime numbers.

$\boxed{7}$ $\boxed{8}$ $\boxed{9}$

A. 3 B. 4 C. 5 D. 6 E. 7

L10 Extention

7. Given n is a certain natural number in $1 \sim 2021$. If the 72 multiple of n is a perfect square, then the maximum value of n is _____ .

8. There are some two-digit numbers, each of which is still a prime number after swapping the first and the second digit. The sum of all these numbers is _____ .

2. Find the number of factors of 196, which is _____ .

L11 extension

1. What is the value of a two-digit number which is 9 times the sum of its digits?

7. The sum of the following numbers in base-10 is ______ .

$(214)_5$ $(ABA)_{16}$ $(1234)_8$

L12 extension

(2) $\frac{4m^2 - 8m + 4}{m^2 - m} =$ ______

A. $\frac{(m-1)}{m}$ B. $\frac{4}{m}$ C. $\frac{4(m+1)}{m}$ D. $\frac{4(m-1)}{m}$

L13 Extention

7 If $a + \frac{1}{b} = 1, b + \frac{1}{c} = 1$, then $c + \frac{1}{a} =$ ______ .

A. 1 B. $\frac{1}{b-1}$ C. $\frac{1}{a(b-1)}$ D. $\frac{1}{a-1}$

6 Solve the following rational equation: $\frac{1}{1-x} - 1 = \frac{2}{x-1}$

$x =$ ______ .

L14

3 Solve the rational equation $\frac{x+1}{x-1} = 1 - \frac{4}{1-x^2}$.

A. $x = 1$ B. $x = -1$ C. $x = \pm 1$ D. No solution

L14 HW

6

(2) $\frac{x+1}{4x^2-1} = \frac{3}{2x+1} - \frac{4}{4x-2}$

$x =$ ______ .

3 If A and B are two constants and $\frac{3x}{x^2-7x+12} = \frac{A}{x-3} + \frac{B}{x-4}$ ($x \neq 3$ and $x \neq 4$), then $A =$ ______ and $B =$ ______ .

5 Apply the Quadratic Formula to solve the following problem and verify your answer by Completing the Square.

There is a positive integer n such that $(n+3)! + 3(n+2)! = (n+1)! \cdot 21$. The value of n is ______ . (Adapted from 2019 AMC 10B Problems, Question #6)

7 If the quadratic equation $(k+1)x^2 + 3x + 1 = 0$ in terms of x has two different solutions, the range of k is ______ .

9 If the quadratic equation $x^2 + 2x - 2m = 0$ in terms of x has two different solutions, the range of m is ______ .

10 The sum of the reciprocals of the roots of the equation $2020x^2 + 2019x + 2018 = 0$ is ______ . (Adapted from 2003 AMC 10A Problems, Question #18)

L17

2. A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible? (2012 AMC 10B Problem, Question #11)

A. 729 B. 972 C. 1024 D. 2187 E. 2304

5. The number 2013 has the property that its units digit is the sum of its other digits, that is $2 + 0 + 1 = 3$. How many integers are less than 2013 but greater than 1000 share this property?

A. 33 B. 34 C. 45 D. 46 E. 58

L17 Extention

2. Five friends a, b, c, d, and e gathered in the park and greeted each other with handshakes. It is known that a shook hands 4 times, b shook hands 1 times, c shook hands 3 times, and d shook hands 2 times. So far, e shook hands ______ times.

1. Aaron, Brian, and Claire are in an ice cream shop. There are 8 different flavors, and they would like to order one scoop of each flavor. If two of them get 3 scoops, and the other gets 2 scoops, how many different ways are there to give them the ice cream?

A. 1680 B. 3360 C. 18816 D. 2352 E. 336

7. There are 8 parking spaces and 4 different cars. How many different possible ways to park these cars so that exactly 3 of the empty parking spaces are next to each other?

A. 1680 B. 840 C. 480 D. 360 E. 240

9 If 3 boys, including Jack and Andy, and 3 girls line up in a row, which of the following statements is incorrect?

A. There are 720 different arrangements.
B. The number of arrangements where Jack is positioned at either end is 240.
C The total number of arrangements where Jack and Andy are adjacent is 120.
D The total number of arrangements where boys and girls alternate is 72.

L18 Extention

9 DeepSeek software is a deeply exploratory artificial intelligence software developed in China. Now rearrange the letters of the word DeepSeek. The number of different arrangements where the letters e are not adjacent is _____ .

A. 24 B. 120 C. 576 D. 2880

10 A, B, C, D, E, F form a line, satisfying the conditions that A, B are adjacent, C, D are not adjacent, and E does not stand at the two ends. How many different arrangements are there?

A. 48 B. 96 C. 72 D. 144 E. 288

9 A team has 5 boys, 4 girls. Now we need to pick 3 people from this team to do the cleaning.

(1) If we are to pick at least 2 boys, there are _____ ways of picking.

(2) If we are to pick at least 1 girls, there are _____ ways of picking.